



Faculty of computer science and information technology

Year 2019/2020

Level:1

Course Name: Calculus I

Course Code: MAT102

Sheet No.:1

Semester: Fall 2019

Instructor: Dr. RawhyIsmail

Problem 1 Find the domain of the following functions:

(a) $y = \frac{3x^2 + 1}{x^2 - 4x - 5}$

(b) $y = \frac{x+1}{x^2 - 9}$

(c) $y = x\sqrt{16 - x^2}$

(d) $y = \frac{x+1}{\sqrt{x^2 - 100}}$

✗ (e) $y = \frac{\sqrt{1-x^2}}{x}$

(f) $y = \frac{3x}{x^2 - 1}$

(g) $y = x(x^2 - 2x + 7)$

(h) $y = \frac{3x}{x^2 + 1}$

Problem 2: Find the derivative of the following functions:

(a) $y = 10x^4 - 3\sqrt{x} + \frac{4}{x^3}$

(b) $y = 6x - \frac{1}{4x} + \frac{9}{x^5} - \frac{2}{\sqrt[4]{x}} - 9$

(c) $y = 6 - 10x + \frac{9}{\sqrt[3]{x}}$

(d) $y = (7x^2 - 8x + 1)^{10}$

(e) $y = \sqrt[3]{2 - 5x + 10x^2}$

(f) $y = \frac{1}{\left(3 - 2\sqrt{x} + \frac{1}{x}\right)^4}$

(g) $y = \sqrt[3]{x} \left(\frac{6}{x} - 1\right)^{10}$

(h) $y = \frac{\sqrt{x} - 1}{\sqrt{x} + 1}$

(i) $y = \sqrt{\frac{x^2}{5x^3 - 4x + 1}}$

✗ (j) $y = \frac{1}{\sqrt{1-x}} \cdot \frac{x-1}{x+1}$

Problem 3 Find the following limits :

(a) $\lim_{x \rightarrow 2} \frac{x}{x^2 - 1}$

(b) $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$

(c) $\lim_{x \rightarrow 1^+} \frac{x}{x - 1}$

(d) $\lim_{x \rightarrow 1^-} \frac{x}{x - 1}$

e) $\lim_{x \rightarrow \infty} \frac{10}{x - 2}$

f) $\lim_{x \rightarrow 0^+} x\sqrt{x}$
